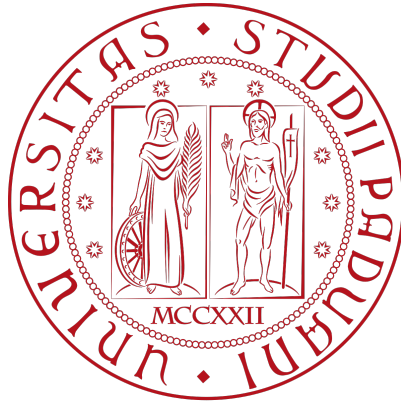




UNIVERSITÀ DEGLI STUDI DI PADOVA
DIPARTIMENTO DI INGEGNERIA INDUSTRIALE

Stability analysis of the gyroscope

*Project of the course "Modeling and simulation of
mechanical systems"*

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Academic year:

2025/2026

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Chapter 1

Introduction

A mechanical gyroscope is a device that employs a rapidly spinning rotor to detect or maintain orientation, based on the principle of angular momentum. The rotor is mounted within gimbals, which allow it to rotate freely along multiple axes. Due to the rigidity of the spinning rotor in space, it resists changes to its orientation, thereby providing stability.

The primary components of a mechanical gyroscope include the rotor or wheel, the gimbals that allow multi-axis rotation, and the supporting frame.

Mechanical gyroscopes have historically been employed in navigation systems for ships, submarines, and aircraft.



Figure 1.1: Simple model of a gyroscope

1.1 Aim of the project

The aim of this project is to analyse the stability of a spinning gyroscope to explain why it is used in many flight instruments.

The work is divided in two parts:

1. **Rigidity in space:** “Gyroscope rigidity in space is the property of a spinning gyroscope to maintain a fixed orientation regardless of any movement of its mounting”.

Several movements of the frame have been applied to analyse the response of the spinning wheel to these disturbances. Afterwards parametric studies have been made to understand what influences most the stability.

2. **Precession:** “Precession is the slow, continuous change in the direction of a gyroscope’s spin axis when a constant torque is applied. Instead of tipping over in the direction of the force, a spinning gyroscope rotates around a third axis, perpendicular to both the angular momentum and the applied torque”.

In order to analyze this effect a force has been applied to create a torque. Different values of the forces have been applied to understand the response of the system. Parametric studies have been made to give a complete view of the system.

Chapter 2

Modeling the geometry of the multi-body system

2.1 Bodies

1. **Frame:** the frame is represented with a torus with a major radius of 40cm and minor radius 1cm. In addition a cylinder has been modeled to represent the outerframe touching the ground. The total calculated mass is about 6.2kg.

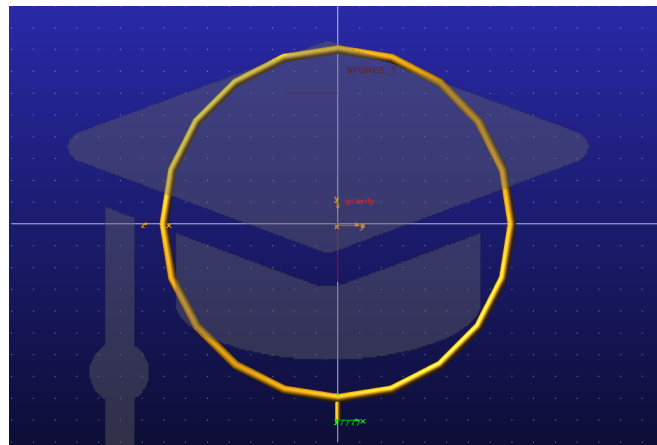


Figure 2.1: Frame in the x-y plane

2. **Outer gimbal:** the outer gimbal is modeled with a torus of major radius 35cm and an inner radius of 1cm. Afterwards it has been attached to the frame with a cylinder of 5cm. Total mass: 5.51 Kg.
3. **Inner gimbal:** the inner gimbal has been modeled with a torus of major

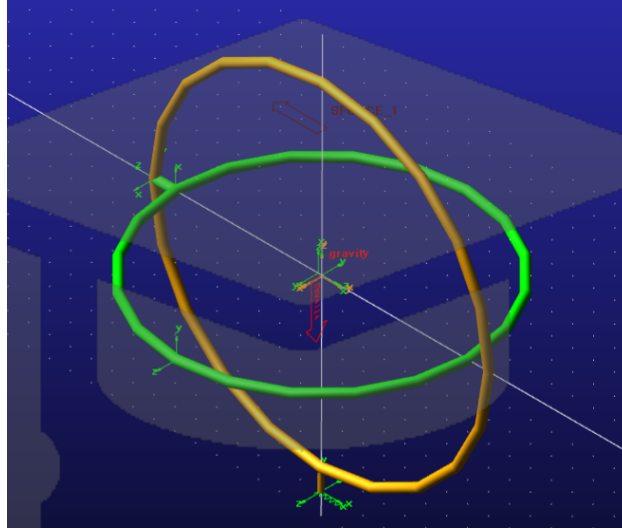


Figure 2.2: Outer gimbal in isometric view (in green)

radius 30cm and a minor radius of 1cm. Later it has been attached to the outer gimbal with a cylinder 5cm long. Total mass: 4.74 Kg.

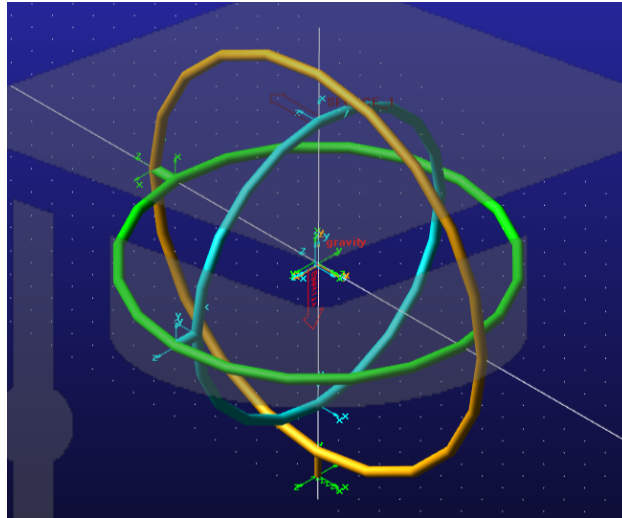


Figure 2.3: Inner gimbal in isometric view (in lightblue)

4. **Inertia wheel:** the spinning axis is represented with a cylinder with a length of 60cm and a radius of 1cm meanwhile the disc has a radius of 28cm and a width of 6cm. Moment of inertia about its x-axis: $4.52E+06 \text{ Kg}\cdot\text{mm}^2$.

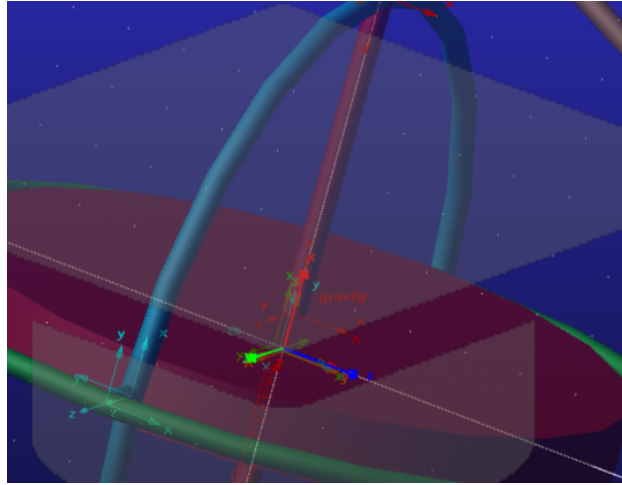


Figure 2.4: Reference frame of the wheel (vertical axis is x)

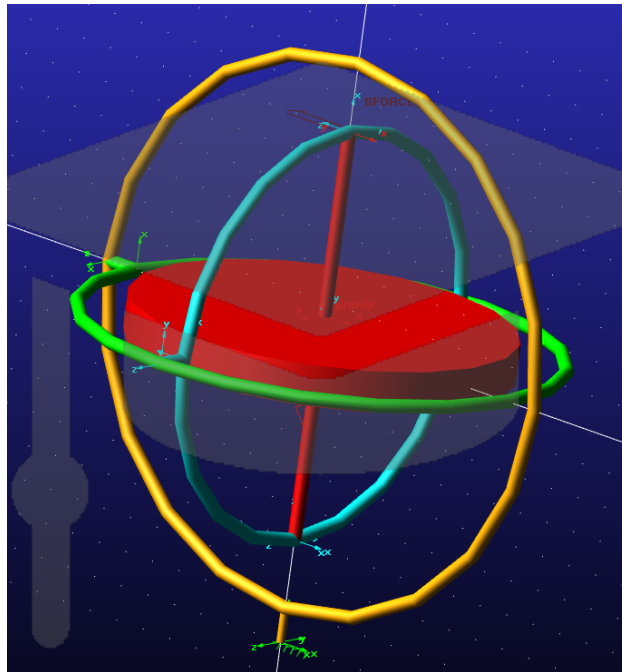


Figure 2.5: Wheel in isometric view (in red)

2.2 Joints

1. **First revolute joint:** the relative motion between the wheel and the inner gimbal has been made with a revolute joint.

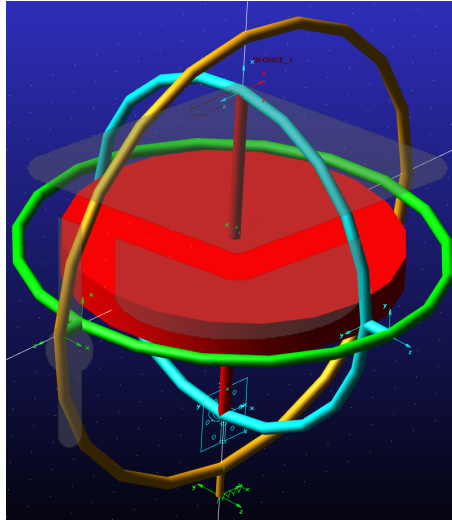


Figure 2.6: Revolute joint between the wheel and the inner gimbal

2. **Second revolute joint:** the relative rotation between the inner gimbal and the outer gimbal is possible with a revolute joint.

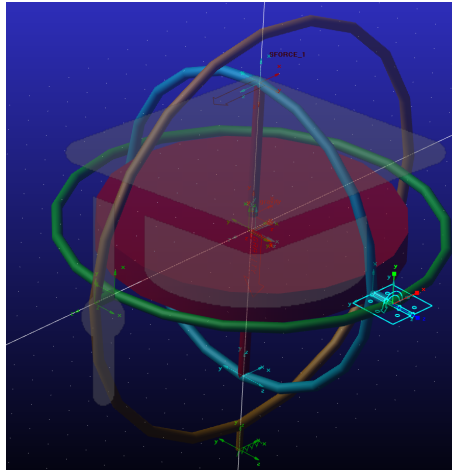


Figure 2.7: Revolute joint between the outer gimbal and the inner gimbal

3. **Third revolute joint:** allows the relative motion between the frame and the outer gimbal.

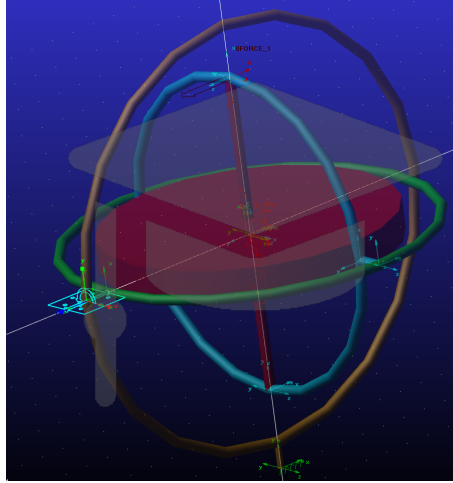


Figure 2.8: Revolute joint between the frame and the outer gimbal

4. **Joint between frame and ground:** spherical joint for moving-frame analysis; fixed joint for precession analysis.

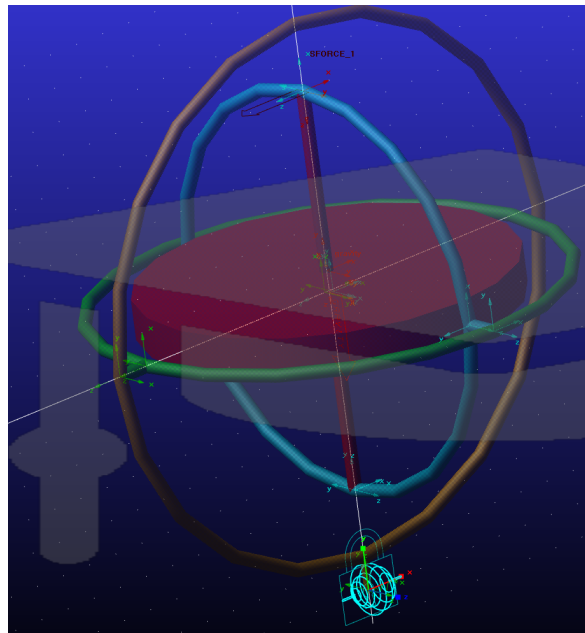


Figure 2.9: Fixed and spherical joints overlapped

2.3 Degrees of freedom

The system without any imposed motion has 4 moving parts, 3 revolute joints and 1 spherical joint. The Gruebler count gives:

$$4 \times 6 - 3 \times 5 - 3 \times 1 = 6 \text{ d.o.f.}$$

Confirmed by Adams:

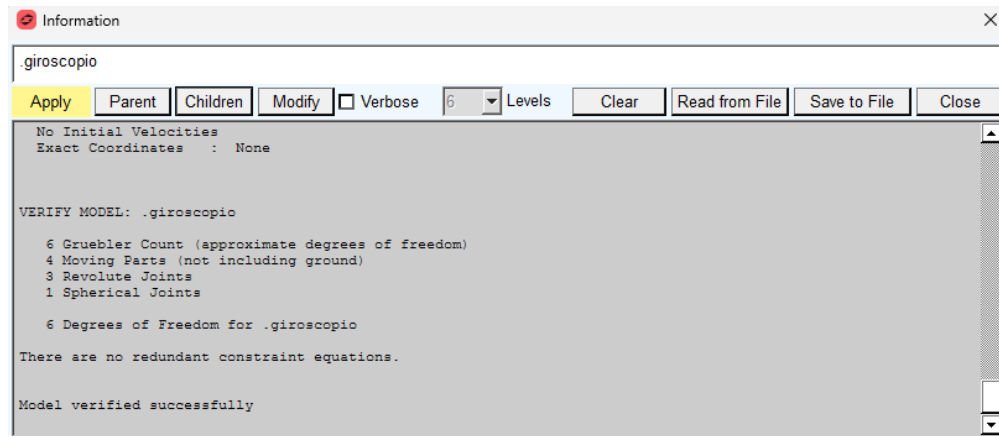


Figure 2.10: Verified model

After imposing motions:

1. **Rotational speed about first revolute joint:** imposed 2000 rad/s.

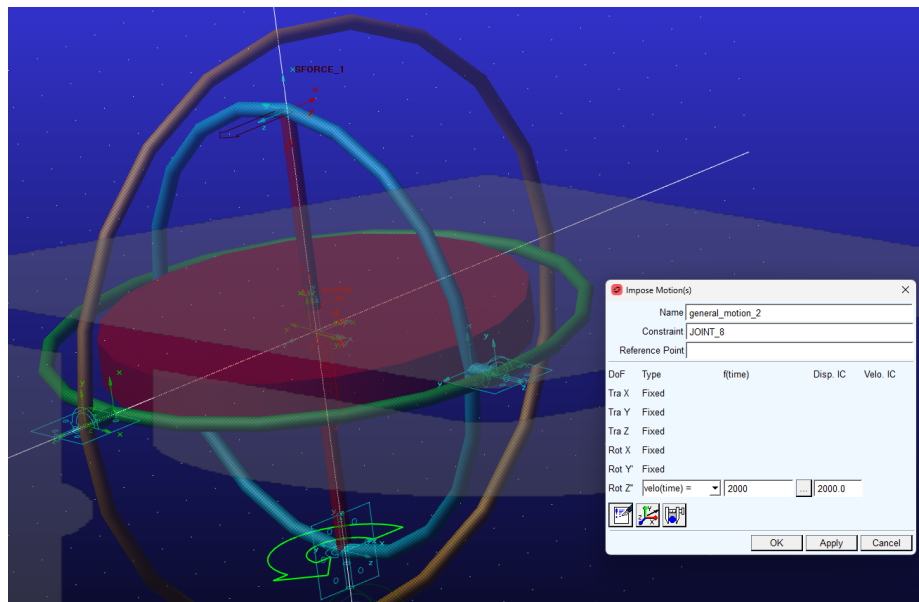


Figure 2.11: Motion imposed to make the wheel spin

2. **Three rotations of the frame:** used step function.

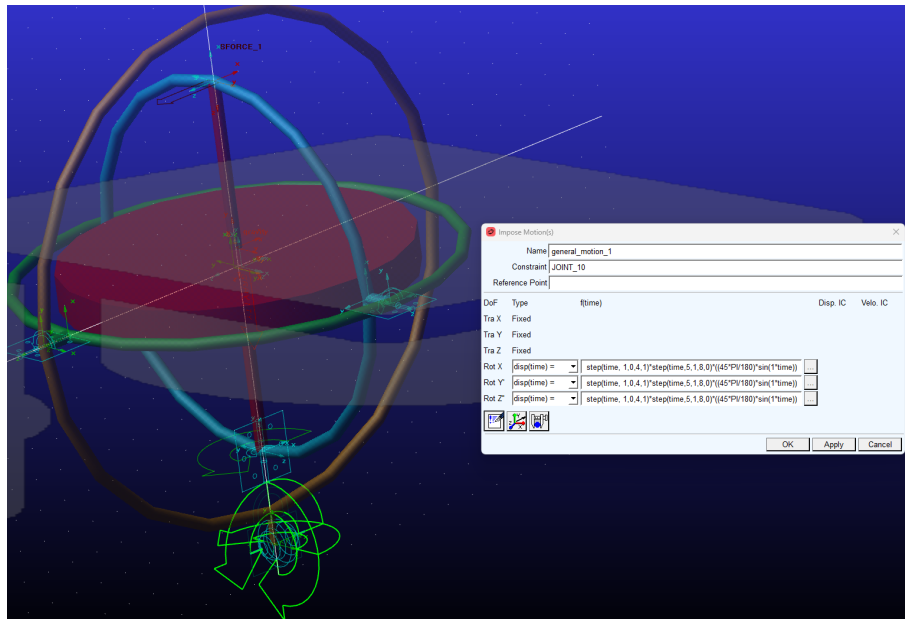


Figure 2.12: Motions imposed to the frame

New d.o.f.:

$$4 \times 6 - 3 \times 5 - 3 \times 1 - 3 - 1 = 2.$$

Confirmed by Adams:

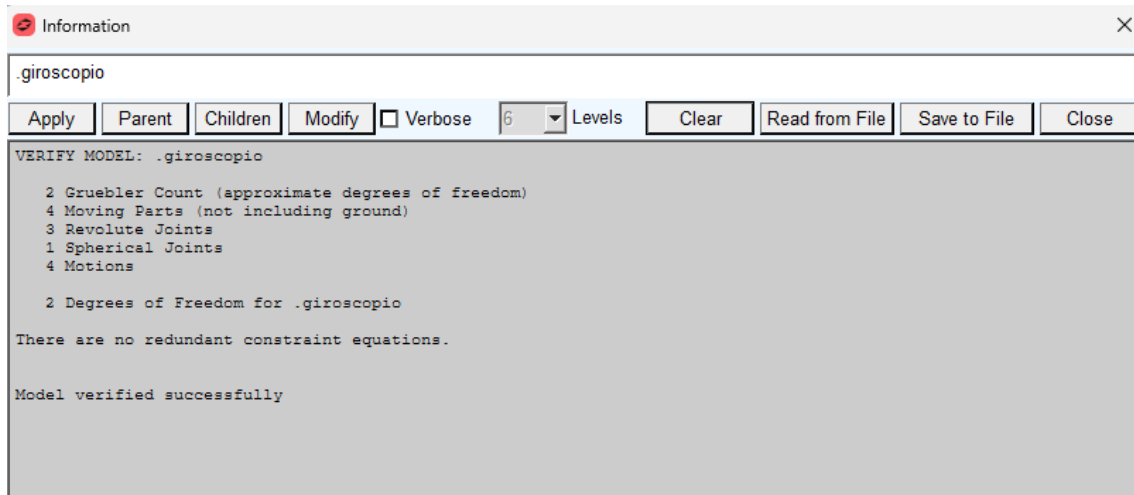


Figure 2.13: Verified model

Chapter 3

Rigidity in space

3.1 Movement of the frame

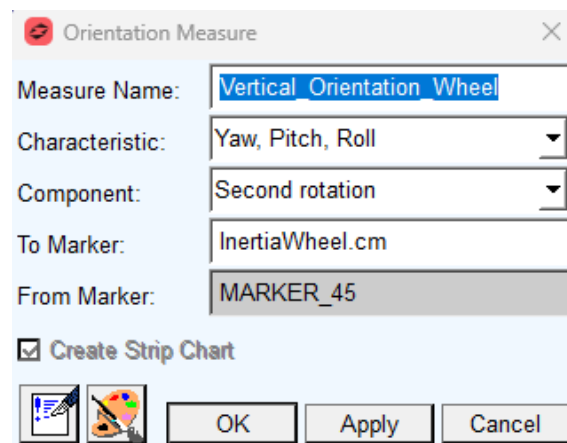


Figure 3.1: Orientation measure (marker45 is a ground marker)

1. One axis rotation

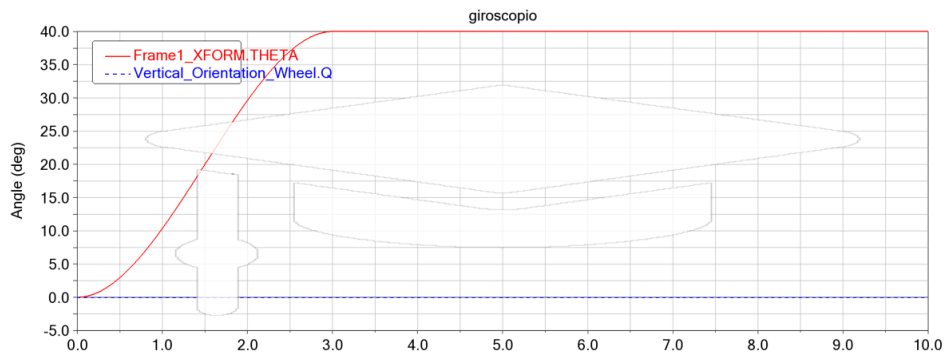


Figure 3.2: Result

2. Instant 40° rotation

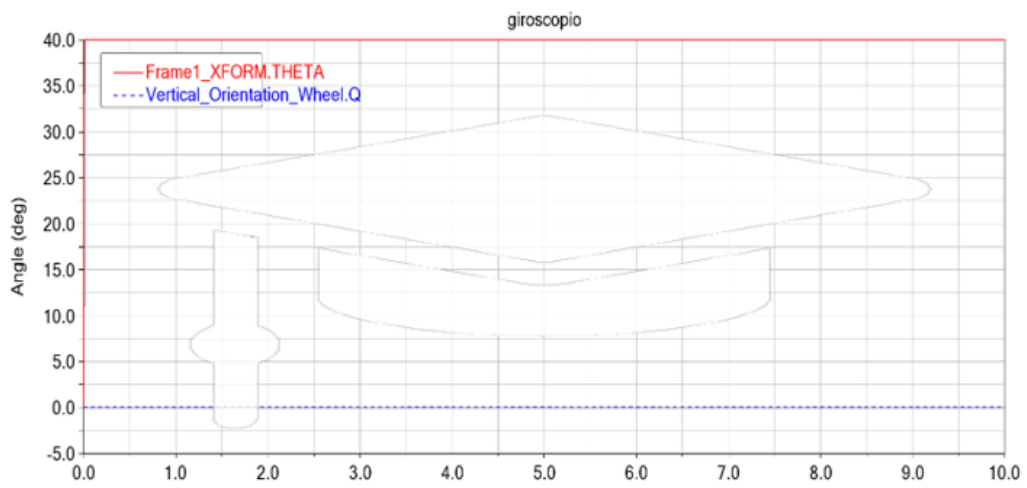


Figure 3.3: Result

3. Three simultaneous rotations

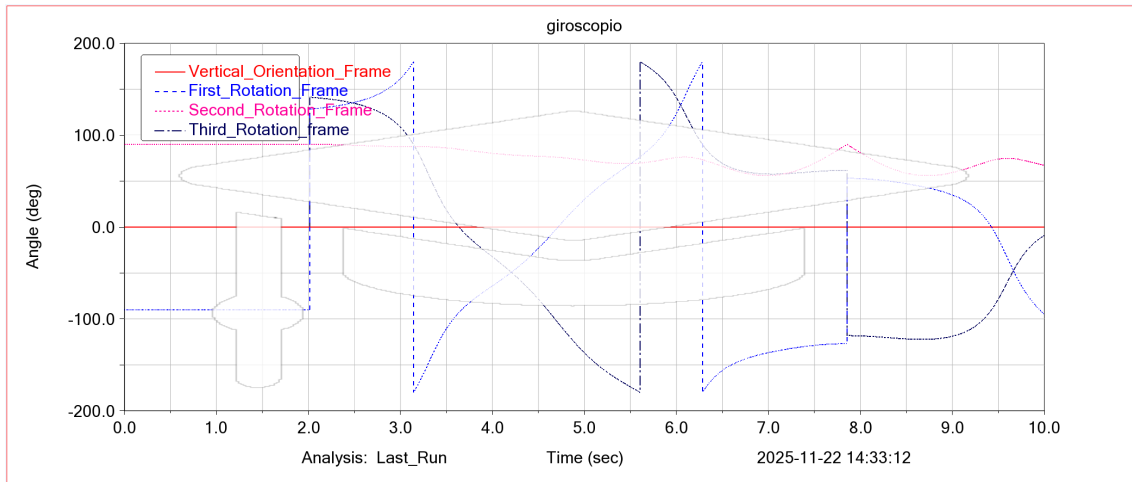


Figure 3.4: Result

3.2 Closer look at vertical orientation

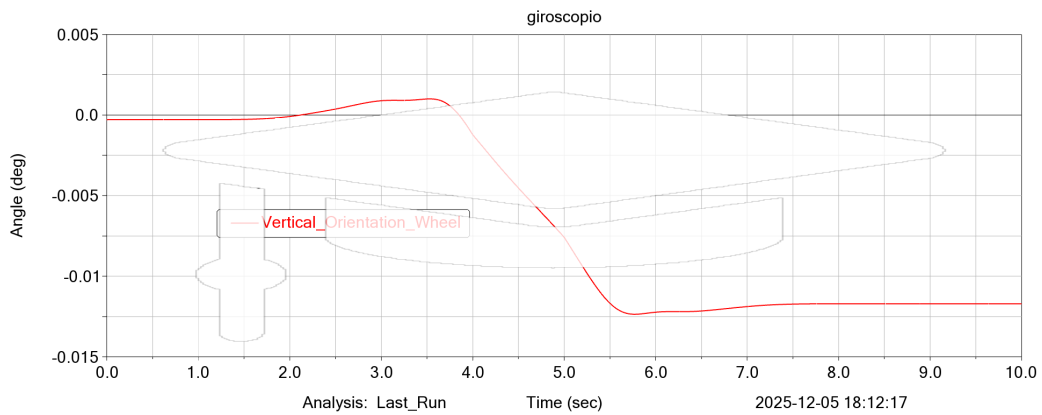


Figure 3.5: Result obtained with the 3-axis rotation

3.2.1 Parametric studies

1. **Changing the rotational speed:** Parametric study of the vertical orientation of the wheel changing the rotational speed while keeping fixed the inertia about the x-axis.

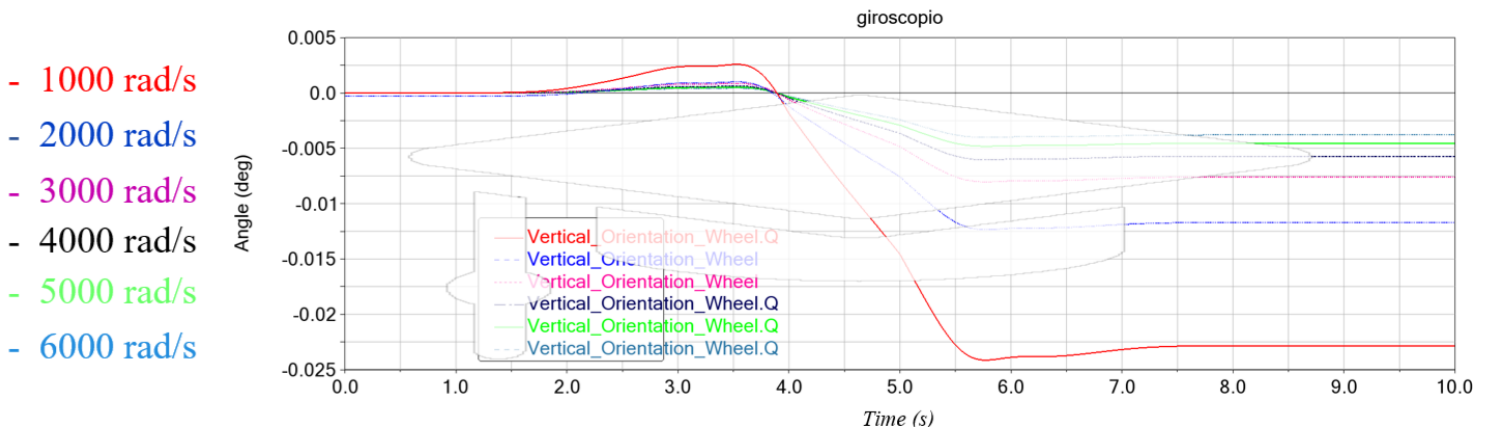


Figure 3.6: Vertical orientation at different values of the inertia about x-axis

From the graph we can tell that the faster the wheel spins the minor is the deviation of the vertical axis: therefore the gyroscope the faster it spins the more precise its measurement is.

2. **Changing the inertia about the x-axis:** to obtain different values of the inertia about the x-axis the width of the disk has been changed.

Wheel width: 60mm
 Moment of inertia about x-axis: $4.52E+06 \text{ Kg*mm}^2$
 Wheel width: 80mm
 Moment of inertia about x-axis: $6.03E+06 \text{ Kg*mm}^2$
 Wheel width: 100mm
 Moment of inertia about x-axis: $7.53E+06 \text{ Kg*mm}^2$
 Wheel width: 120mm
 Moment of inertia about x-axis: $9.04E+06 \text{ Kg*mm}^2$
 Wheel width: 140mm
 Moment of inertia about x-axis: $1.05E+07 \text{ Kg*mm}^2$

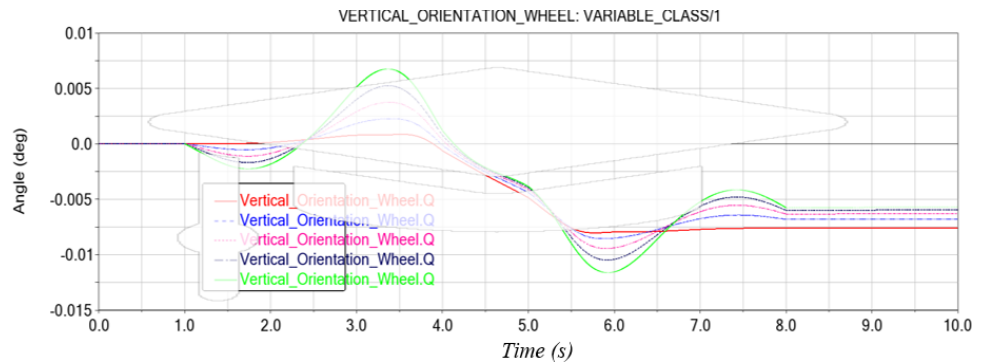


Figure 3.7: Vertical orientation at different values of the inertia about x-axis

The graph can be divided in two parts: transitory and regime. In the transitory fase the higher the inertia the higher is the deviation meanwhile in the regime part the higher is the inertia the lower is the deviation form the vertical axis. Of course the important is the regime fase so the higher the inertia is, the huihger is the accuracy of the measurement.

This graphs have been obtained with this imposed rotations of the frame:

Chapter 4

Precession

“Precession is the slow, continuous change in the direction of a gyroscope’s spin axis when a constant torque is applied. Instead of tipping over in the direction of the force, a spinning gyroscope rotates around a third axis, perpendicular to both the angular momentum and the applied torque”

To make the visualization easier a constant force (1000N) has been applied on the inner gimbal in order to create a torque.

A schematic view of the acting forces is presented:

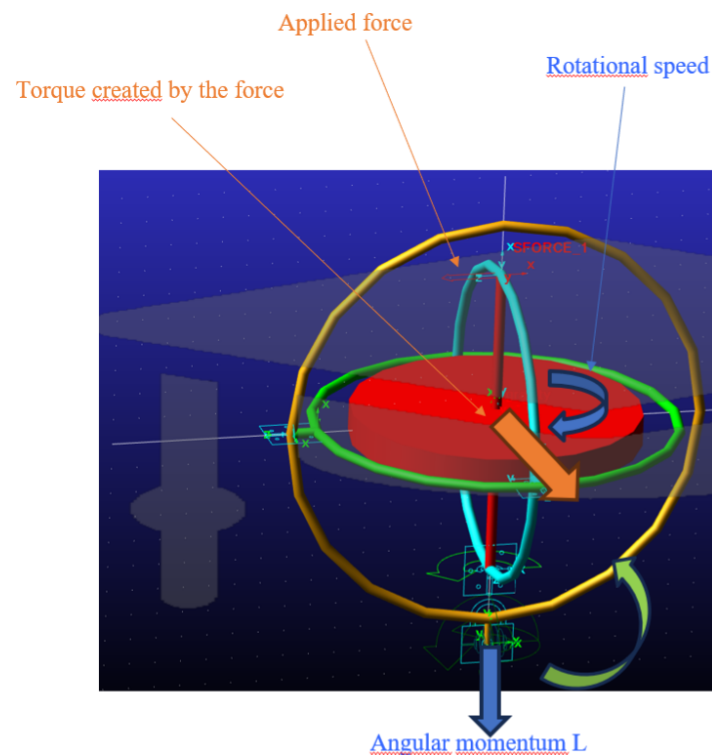


Figure 4.1: The green arrow is the precession of the rotation of the gyroscope

The simulation confirms the forecasted rotation of the gyroscope:

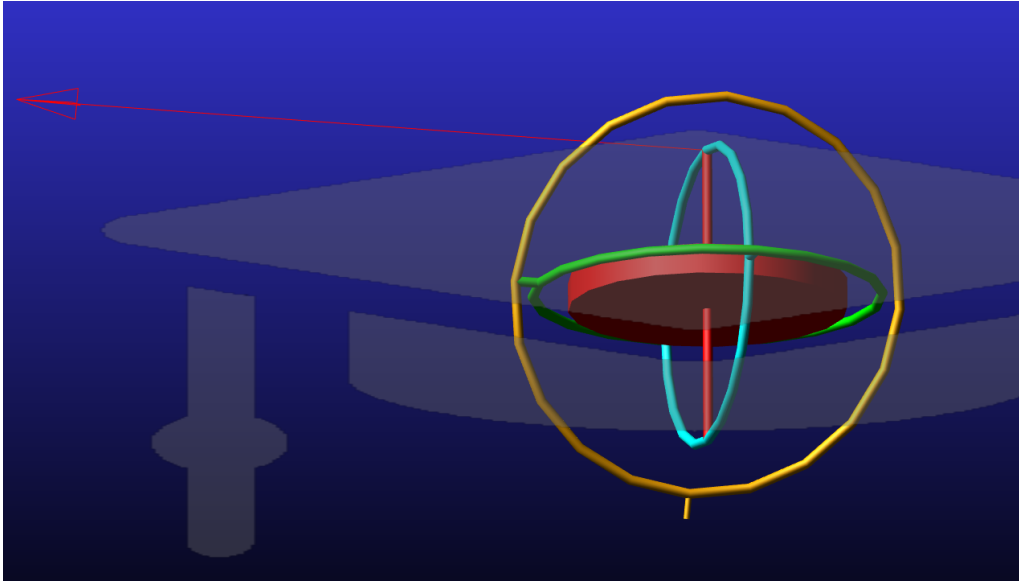


Figure 4.2: The green arrow is the prevision of the rotation of the gyroscope

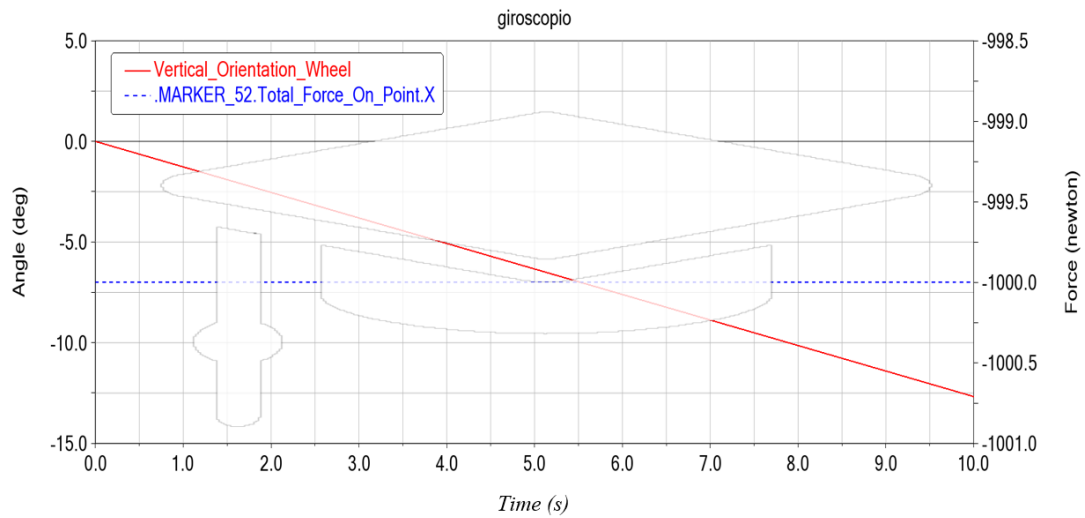


Figure 4.3: Vertical orientation with a constant force applied

4.1 Impulse force

In the next pages the response of the system is analyzed when an impulse force is applied.

On Adams the step function has been used: `step(time,0.1,0,0.15,100)step(time,0.15,100,0.2,0)`. This function simulates an impulse disturbance applied upon inertia wheel that has a width of 0.1 second and an amplitude of 100 N

4.1.1 Parametric studies

Three different parametric studies have been conducted.

1. **Parametric study of the vertical orientation with different values of the impulse force with inertia about x-axis ($4.519\text{E}+06 \text{ Kg}\cdot\text{mm}^2$.) and rotation speed (2000 rad/s) fixed:**

- 100 Newton
- 200 Newton
- 300 Newton
- 400 Newton
- 500 Newton

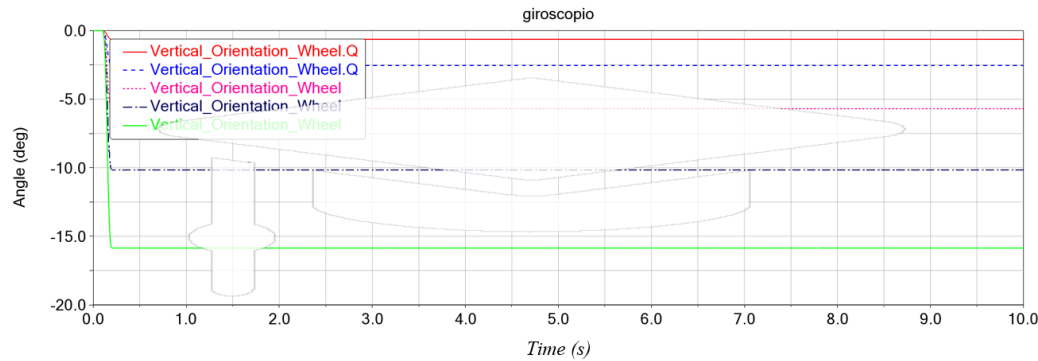


Figure 4.4: Vertical orientation with a constant force applied

The graph shows a simple deduction: the stronger the force the greater is the change in the vertical orientation

2. **Parametric study of the vertical orientation with different values of the inertia about x-axis with speed (2000 rad/s) and impulse force (400 N) fixed:**

This parametric analysis shows that the greater the inertia is, the smaller is the deviation of the vertical axis. Therefore we can say that the greater the inertia is, the more stable the gyroscope is when an impulse disturbance is present.

Wheel width: 60mm
 Moment of inertia about x-axis: $4.52\text{E}+06 \text{ Kg}\cdot\text{mm}^2$
 Wheel width: 80mm
 Moment of inertia about x-axis: $6.03\text{E}+06 \text{ Kg}\cdot\text{mm}^2$
 Wheel width: 100mm
 Moment of inertia about x-axis: $7.53\text{E}+06 \text{ Kg}\cdot\text{mm}^2$
 Wheel width: 120mm
 Moment of inertia about x-axis: $9.04\text{E}+06 \text{ Kg}\cdot\text{mm}^2$
 Wheel width: 140mm
 Moment of inertia about x-axis: $1.05\text{E}+07 \text{ Kg}\cdot\text{mm}^2$

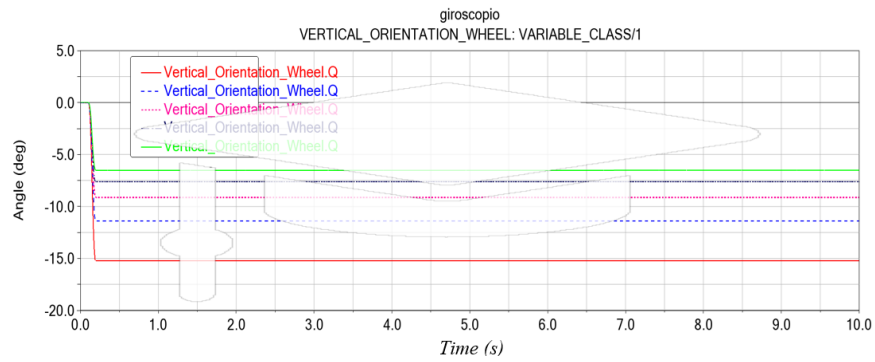


Figure 4.5: Vertical orientation with a constant force applied

3. Parametric study of the vertical orientation of the wheel changing the rotational speed while applying an impulse force of 400N:

As expected, the rotational speed plays a very important role in the stability of the system: the slower it spins the greater is the deviation when an impulse force is applied

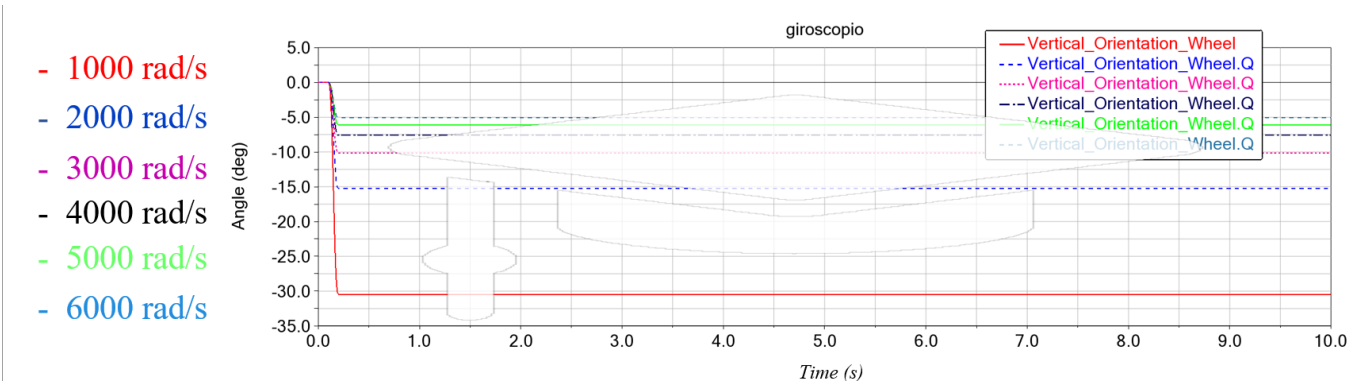


Figure 4.6: Vertical orientation

Chapter 5

Conclusion

AFTER THE CONDUCTED PARAMETRIC STUDIES, IT CAN BE CONFIRMED THAT THE FASTER A GYROSCOPE SPINS AND THE GREATER ITS INERTIA, THE MORE STABLE AND RELIABLE ITS MEASUREMENTS BECOME.